

Manufactured Skill in Prediction-Market Forecasting

A two-venue trajectory benchmark, and the preprocessing step that creates apparent predictability on it

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Abstract

Prediction-market probability paths are an appealing forecasting target: dense, continuous, bounded, and economically interpretable. No reproducible, market-disjoint benchmark for them exists publicly. We build one — PARITY-U — comprising 100,000 held-out probability trajectories from resolved markets on *two* independent venues, Polymarket and Kalshi, with deterministic hash-based selection, family-disjoint splits, and released manifests.

Constructing it surfaced a result we consider more durable than any accuracy number. Dropping consecutive duplicate prices — the natural implementation of “keep only price-change events” — yields a benchmark on which the persistence baseline is wrong at *every* step by construction. Its direction accuracy is then necessarily 0.000, and its RMSE is inflated by 27.4% on a population-matched comparison, so any learned model scored against it is flattered by that amount before it learns anything. The effect replicates across venues to within 0.6 percentage points, and the step is also commonly implemented incorrectly: exact floating-point deduplication leaves 11.7% of “price changes” below a twenty-millionth of a tick, which *suppresses* the very artifact it creates.

On the honest grid a small residual Transformer does beat persistence significantly on both venues and both grids, every seed’s bootstrap interval excluding zero. But the gain is narrow rather than general: confined to the volatile tercile (−2.25%), *worse* than persistence on quiescent markets (+7.50%), decaying with horizon against our pre-registered expectation, and absent in MAE. Priced in the units a trader faces, the entire directional edge sustains a round-trip cost of 0.0014 \$/share — about one seventh of a tick — against spreads of one to two ticks. Predictability here is real, narrow, and not tradeable. We release both time grids, the family-disjoint builder, and *parity-check*, a diagnostic that reports the inflation for any irregularly sampled series.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Information systems** → *Data mining*; • **Applied computing** → *Economics*.

Keywords

prediction markets, benchmark construction, time-series forecasting, data leakage, market microstructure, evaluation

1 Introduction

A prediction market quotes a continuously updated probability for a well-defined future event. The resulting price path is an unusually clean forecasting target: it is bounded in $[0, 1]$, it is dense in time, it aggregates dispersed private information [15], and — unlike most

financial series — its units are directly interpretable, since a contract settling in $\{0, 1\}$ makes a probability unit exactly one dollar per share.

Despite this, there is no public, reproducible benchmark for forecasting prediction-market probability *trajectories*. The empirical literature concentrates on whether market prices predict *outcomes* well [1, 10, 13], which is a different question with a different target. Work that does forecast the paths tends to use ad hoc extractions whose selection, splitting, and time-indexing choices are not recoverable from the paper.

Those choices turn out to matter more than the model. This paper’s central finding is a construction artifact:

Dropping consecutive duplicate prices — the standard way to convert an irregularly sampled feed into a “price-change event” series — produces a dataset on which the persistence baseline is wrong at every step by construction. Its direction accuracy is then exactly 0.000 as a matter of arithmetic, and its RMSE is inflated by 27.4%. Every learned model evaluated on such a grid is credited with skill it does not have.

This is arithmetic, not inference. If consecutive duplicates are removed then $y_t \neq y_{t-1}$ holds by construction, so a forecast of “no change” is necessarily wrong at every step. No choice of model, seed, or bootstrap can change it. We quantify the magnitude, show it replicates across two independently constructed venues, and show that the obvious implementation of the step is additionally buggy in a way that *understates* the artifact.

Contributions. We release **PARITY-U**, a reproducible two-venue benchmark of 100,000 probability trajectories per grid, with deterministic hash-based selection, family-disjoint splitting, released manifests, and *both* time grids shipped so the artifact can be measured rather than inherited (Section 3). We quantify the deduplication artifact at +27.4% persistence RMSE inflation on a matched population, replicating to within 0.6 pp on a second venue, and show that exact-float deduplication leaves 11.7% of retained “events” economically identical (Section 4). We show the standard direction-accuracy metric is structurally uninformative on *both* grids — identically 0.000 on one, identically the flat-step fraction on the other — and give the conditioning that repairs it (Section 5). Finally we report a careful positive result and its boundaries (Sections 8–9), and release *parity-check*, a portable diagnostic for grid-induced baseline inflation (Section 4.4).

We consider the negative results in this paper to be its most reusable content. A benchmark that other people download is worth less than a check other people run.

2 Related work

Prediction markets. The economic literature establishes that market prices aggregate information and forecast outcomes competitively with polls and expert panels [1, 13, 15], and studies the interpretation of the price as a probability [10, 16]. That literature evaluates *terminal accuracy*. We take the complementary target: the short-horizon evolution of the price path itself, treated as a forecasting problem, with no reference to the realised outcome. We deliberately report no Brier score, since our targets are continuous probabilities rather than realised binary events.

Forecasting baselines and leakage. A recurring lesson in the forecasting literature is that simple baselines are hard to beat and are frequently mis-specified in comparisons [5, 9], and that deep architectures have repeatedly been found to underperform trivial baselines once evaluation is tightened [17] — a critique that runs alongside the steady elaboration of Transformer forecasters themselves [11, 18]. Our result strengthens that lesson in a specific direction: here the baseline is not merely strong, it is *structurally disabled* by a preprocessing convention, invisibly unless measured. Separately, data leakage is a recognised cause of overstated results [6, 7], and validation design for temporally or hierarchically structured data is well studied [2, 12]. Our leakage channel is hierarchical rather than temporal: distinct contracts on one underlying event are separate markets that hash independently, so a market-ID split leaves same-event siblings on both sides (Section 6).

Machine learning in finance. Gu et al. [4] establish the value of flexible learners for asset-pricing prediction while emphasising careful out-of-sample protocol, and López de Prado [8] catalogues the failure modes — overlapping samples, multiple testing, backtest overfitting — that make financial ML results fragile. Our protocol adopts the corresponding defenses: splits are a pure function of the family key, configuration selection touches the validation split only, the test split is read exactly once, and significance is reported per seed rather than pooled.

What is new here. To our knowledge no prior work quantifies the effect of the event-grid convention on baseline strength in prediction markets, and no public trajectory benchmark for these markets exists to quantify it against.

3 The PARITY-U benchmark

3.1 Task

Given the last $C = 30$ observed probabilities of a market outcome, forecast the next $H = 10$. We treat the probability path as the observable of a partially observed dynamical system and ask whether its short-horizon evolution carries *incremental* predictive value over well-specified non-learned dynamics.

This framing is deliberately modest. We do not claim a latent world model, an agentic simulator, or counterfactual capability.

3.2 Construction

The source is resolved markets from Polymarket (2,161 price shards) and Kalshi. Market status == resolved is the only market-side filter; the resolution itself is never used as a feature.

Selection. We take one 40-observation window per market outcome. Outcomes are retained by a stable hash of `platform_market_id :: outcome_id`, pushed into the lazy query *before* collection. This bounds memory and — unlike taking the first N groups in shard or lexicographic order — yields a representative sample across all shards. The retained band is then trimmed deterministically to exactly 100,000 windows in hash order. The window offset within an outcome is itself derived from the outcome hash, so it does not depend on group iteration order.

Splitting. An *independent* hash assigns 70/15/15 train/validation/test. Section 6 discusses the split key in detail; the released default groups every contract on one underlying event into the same split.

Table 1: The two released grids (Polymarket). Both contain 100,000 windows and differ only in how time is indexed: the event grid drops consecutive duplicate prices, the tick grid retains raw ticks.

Variant	Train	Validation	Test
Event (default)	70,264	14,849	14,887
Tick (robustness)	70,309	14,812	14,879

Two time grids, both released. The grid choice is not cosmetic. It is the single most consequential construction decision in this benchmark, and it is the subject of the next section.

4 Deduplication manufactures skill

4.1 The mechanism

Prediction-market feeds are irregularly sampled and mostly quiescent: long stretches pass with no trade and no quote revision. The natural cleanup is to keep only price-*change* events, i.e. drop consecutive duplicates. This is reasonable on its face — it removes uninformative repetition and shortens sequences.

It also guarantees $y_t \neq y_{t-1}$ for every t . The persistence forecast $\hat{y}_t = y_{t-1}$ is therefore wrong at *every* step, and its direction accuracy is exactly 0.000 — not empirically, but as a consequence of the filter. Table 2 shows what this does to the primary baseline.

Table 2: The event grid disables the persistence baseline. Test split, both venues, deduplication-corrected (Section 4.3).

Venue	Grid	n_{test}	zero- Δ	Pers. RMSE	Pers. DirAcc
Polymarket	event	14,887	0.00%	0.10604	0.0000
Polymarket	tick	14,879	45.25%	0.09622	0.4477
Kalshi	event	10,229	0.00%	0.10996	0.0000
Kalshi	tick	12,755	42.81%	0.09928	0.4222

4.2 The magnitude, on a matched population

The as-built comparison in Table 2 gives a persistence RMSE inflation of +10.20% on Polymarket and +10.76% on Kalshi. Those two figures are directly comparable to each other and agree to within 0.6 percentage points on two independently constructed datasets.

They also carry a confound, which we correct. The event grid requires ≥ 40 observations *after* deduplication; the tick grid requires ≥ 40 before it. Since deduplication can only shorten a series, the event grid’s eligible population is a strict *subset* of the tick grid’s — 245,059 against 321,021 candidate outcomes — and after each is trimmed to 100,000 the two selected sets overlap only 76.4%. The as-built figure therefore mixes the effect of re-indexing time with the effect of changing which markets qualify.

Restricting both grids to the 76,387 outcomes present in each isolates the former:

Table 3: Holding the population fixed nearly triples the measured artifact. Polymarket, test split.

	Event RMSE	Tick RMSE	Inflation
As built (populations differ)	0.10604	0.09622	+10.20%
Matched (76,387 series)	0.10530	0.08268	+27.36%

Both numbers are correct answers to different questions. The as-built figure characterises the two grids *as released*; the matched figure is the **causal effect of the deduplication step**, which is the quantity this paper’s central claim concerns. We report the matched figure as the headline with the as-built figure alongside.

One caveat travels with it, and cannot be removed by matching: matching fixes the *series* but not the *window*. The window origin is a function of series length and the deduplicated series is shorter, so the two grids read different spans of the same market. This is intrinsic to comparing two time indexings.

4.3 The step is also commonly implemented incorrectly

The natural implementation of “drop consecutive duplicates” is an exact inequality on the stored array. That is insufficient, and the failure is invisible unless looked for.

The Polymarket feed is float64 and its values arrive marginally off-grid: the quoted price 0.45 is delivered as 0.449999958... Casting to float32 before deduplicating maps one economic price onto two adjacent float32 values one ULP apart (2.98×10^{-8} near 0.45), and an exact-equality filter retains both as a “price change”. Measured on our event grid before correction:

Table 4: Retained “price-change events” that are not economically price changes. Event grid, test split, before correction.

$ \text{price move} \leq$	share of steps
exactly 0	0.00%
1×10^{-7}	7.58%
1×10^{-6}	11.74%
1×10^{-3}	16.45%

The smallest non-zero move was 4.7×10^{-10} , against a *modal* real price increment of 0.0100 — roughly twenty million times smaller than one tick. So 8–12% of nominal price-change events were not price changes, and the grid’s defining property held only under exact float equality rather than economically.

We correct this by snapping to the quote grid (rounding at 10^{-6} , a $250\times$ margin over the smallest genuine inter-quote gap of 2.5×10^{-4}) *before* deduplicating. Residual sub- 10^{-6} steps fall from 11.56% to 0.06%.

The correction strengthens the finding. The measured artifact rises from +6.58% to +10.20% as built: the implementation defect was *suppressing* the very effect the benchmark exists to demonstrate. It also explains an apparent venue difference. Before correction, Polymarket measured +6.58% against Kalshi’s +10.76%, which read as a genuine cross-venue disagreement. It was not — Kalshi’s feed is already on-grid, so its arrays are byte-identical before and after the fix, and correcting Polymarket makes the two converge.

Recommendation. Deduplicate on a tolerance of one tick, never on floating-point equality — and verify rather than assume, since the defect is feed-specific.

4.4 A portable check

The artifact is not specific to prediction markets: any irregularly sampled series converted to a change-event index inherits it. We therefore release parity-check, a standalone script that reports, for any price series, the share of nominal “changes” below one tick, the smallest non-zero move, persistence RMSE under both indexings, and the resulting inflation. It reproduces the +10.20% and +10.76% figures above from the released grids. Two guards are built in, both from our own mistakes: it computes the sub-tick share among *non-zero* moves only, and it refuses to report an inflation figure when the derived event grid retains under 50% of the input series.

Three properties make this the paper’s most durable claim. It is arithmetic rather than inference, so it survives any change to the model. It attaches a number to a preprocessing choice that looks innocuous and is widely made. And it is a property of the *step*, not of a venue.

5 The direction metric is uninformative on both grids

The event grid forces persistence to 0.000. The tick grid has the mirror-image defect: persistence scores *exactly* the flat-step fraction, because it is right on every step where nothing happens and wrong on every step that moves. On the Polymarket tick test split, 44.77% of steps do not move and persistence scores 0.4477 — agreement to four decimals.

The metric is therefore an artifact of grid construction on both grids, and carries zero directional information about the baseline in either case. Conditioning on realised movement repairs it at no cost:

Conditional on movement, the learned model is the only model meaningfully above zero — but only +1.03 pp (tick) and +2.00 pp (event) above a coin flip. That supports no tradeable claim, as Section 9 makes precise.

Reporting the unconditional metric alone credits persistence with skill it does not have. Both numbers should be reported.

Table 5: Direction accuracy, Polymarket tick grid, test split. The unconditional column describes the benchmark; the conditional column describes the model.

Model	all steps	steps that move	flat steps
Persistence	0.4477	0.0000	1.0000
Moving average	0.4457	0.0551	—
Exponential smoothing	0.3524	0.4975	—
Residual Transformer	0.2979	0.5103	—

6 Event-family leakage

A hash of the market ID gives no protection against *event*-level leakage. Different contracts on the same game or the same city-day are separate markets and hash independently, so a temperature ladder can have its 65° bin in train and its 66° bin in test.

The exposure is large. Across the resolved Polymarket corpus, 57.7% of markets belong to a multi-market event family and 88.9% of those families straddle a split. Within the benchmark itself:

Table 6: Test windows whose event family also appears in training. The family split hashes an event key rather than the market ID.

Venue	Grid	Market split	Family split
Polymarket	event	19.86%	0.00%
Polymarket	tick	24.02%	0.00%
Kalshi	event	39.14%	0.00%
Kalshi	tick	45.19%	0.00%

Two observations are worth recording for anyone constructing a similar benchmark.

Population statistics do not predict benchmark exposure. At the population level Kalshi looks *less* exposed than Polymarket — its families straddle at 62.2% against 88.9%, because they hold 5.5 members on average against 12.5. That ordering *reverses* at benchmark level, because the ≥ 40 -observation requirement selects Kalshi’s liquid, actively quoted markets, and those are precisely the ones sitting in large ladder families. The eligibility filter concentrates family membership on one venue and thins it on the other. Population rate is not a proxy for benchmark exposure in either direction, and we were misled by it before measuring directly.

The event key must be structural where possible. On Kalshi, tickers are structurally SERIES-EVENT-STRIKE, so the event key is the first two segments and parses at 100.0%. On Polymarket we must recover it from the slug with a date-token regex, which parses 65.4%; unparsed slugs become singleton families, which can only *undergroup* and so leave the split no worse than a market-ID split. This is why we claim **family-disjointness** and never complete event-disjointness. Families of the form <asset>-above-<strike>-on-<date> are a known residual, since the strike precedes the date and does not group.

What we do and do not establish. The family-disjoint builder is released and is the benchmark’s default, and it drives measured

exposure to 0.00% on both venues. We do *not* yet report how much the previously leaked signal was worth. Measuring that requires retraining both arms under a single query plan and comparing them on the windows they share, and a paired comparison we attempted was confounded by an unrelated tie-ordering non-determinism in the builder (Section 10). We therefore report every model result in this paper under the **market-ID split** — the configuration in which the full seed matrix was trained and independently replicated — and treat the effect of the stricter split as an open quantity rather than asserting a number for it.

7 Models and protocol

7.1 Models

All models are evaluated on identical held-out trajectories and horizon points.

- **Persistence** — repeat the last observed probability. The primary baseline. Market probabilities are highly persistent and this is hard to beat.
- **Moving average** — mean of the last 5 observations.
- **Exponential smoothing (Holt)** — level plus trend, extrapolated over the horizon and clamped to $[0, 1]$. Vectorised and deterministic, run on the *full* test set, so there is no evaluation-population mismatch to disclose. ARIMA was deliberately excluded: per-trajectory fitting at 100k scale would have forced exactly such a mismatch.
- **Residual Transformer** (proposed) — scalar price embedding, sinusoidal positional encoding, Transformer encoder [14], MLP head predicting a *bounded adjustment* to the last observed probability, output clipped to $[0, 1]$. This makes persistence a nested special case and cannot emit invalid probabilities.
- **Free Transformer** (ablation) — identical, but predicts the path directly rather than a residual. It exists to answer the obvious question: is the gain merely the persistence anchor?

7.2 Protocol

Configuration selection used the validation split only; the test split is read exactly once, by the final evaluation. The sweep gave every configuration an equal 40-epoch budget at seed 7, batch 4,096.

Two protocol facts are disclosed rather than smoothed over. *Early stopping never fired at the sweep budget* — every configuration was still improving at epoch 40 with a near-zero train/validation gap, so these models underfit rather than overfit, and selection is “best under an equal budget” rather than “best converged”. And *the grid was extended after the fact*: the base grid was $d_{\text{model}} \in \{64, 128\} \times \text{layers} \in \{2, 3\} \times \text{lr} \in \{3\text{e-}4, 1\text{e-}3\}$, whose winner sat on the edge, so we probed past it ($d = 256$, $\text{lr} = 3\text{e-}3$, layers = 4) on validation only. Every probe came back worse, so the selected configuration is *interior in all three swept dimensions*.

Final runs use the frozen configuration $d_{\text{model}} = 128$, layers = 3, lr = 1e-3 (validation MSE 0.0103531) trained to convergence (140 epochs, patience 12) for 5 seeds. The ablation uses 3 seeds at the same budget. The tick-grid variant retrains the same frozen configuration for 3 seeds without re-selecting it, so that variant is

a genuine transfer-of-protocol check rather than a second bite at selection.

7.3 Metrics and uncertainty

RMSE is primary. MAE is reported even where it disagrees. Direction accuracy is successive-increment sign agreement, with the last observed probability as the pre-horizon anchor; ties are exact-sign matches, so a flat forecast scores only where the market is also flat. This definition was frozen before the final runs and applied identically to every model.

Uncertainty is a paired bootstrap over held-out *trajectories* — not horizon points, which are correlated within a trajectory — with 2,000 draws, computing the paired RMSE difference against persistence on each draw.

Two reporting rules we hold ourselves to:

- Learned rows are the **across-seed mean of per-seed metrics**, i.e. typical single-model performance, never the metric of an ensembled prediction. The seed ensemble appears only as its own explicitly labelled row.
- Significance is **per seed**. We report whether *every* individual seed’s interval excludes zero, and we never claim an improvement on the strength of a pooled or ensembled interval.

Pre-registered gate. An improvement is claimed only if every seed is significant on the event grid *and* the same direction survives on the tick grid, where persistence is genuinely strong.

8 Results

8.1 Main comparison

Table 7: Event grid, Polymarket, $n_{\text{test}} = 14,887$. * = every individual seed’s CI excludes zero.

Model	RMSE	MAE	DirAcc	dRMSE vs pers.
Persistence	0.10604	0.05921	0.0000	—
Moving average	0.11164	0.06642	0.0579	+0.00561
Exp. smoothing	0.11413	0.06471	0.4966	+0.00809
Free Transformer	0.10425	0.05954	0.5187	−0.00179*
Residual Tr.	0.10408	0.05919	0.5200	−0.00196*

Table 8: Tick grid, Polymarket, $n_{\text{test}} = 14,879$.

Model	RMSE	MAE	DirAcc	dRMSE vs pers.
Persistence	0.09622	0.04488	0.4477	—
Moving average	0.10264	0.05196	0.4457	+0.00642
Exp. smoothing	0.10336	0.05051	0.3524	+0.00714
Residual Tr.	0.09520	0.04550	0.2979	−0.00102*

Bootstrap intervals for the learned rows are $[-0.00255, -0.00138]$ (event, residual), $[-0.00233, -0.00126]$ (event, free) and $[-0.00158, -0.00050]$ (tick). Seed standard deviations are ± 0.00006 (event residual), ± 0.00004 (event free) and ± 0.00006 (tick). 100% of bootstrap draws favour the learned model on both grids, and the pre-registered gate passes.

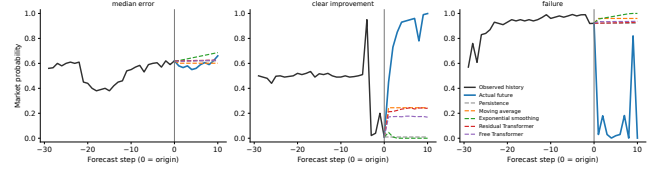


Figure 1: Representative held-out trajectories with the residual model’s forecast against persistence. The model tracks the anchor closely and deviates mainly where the path is volatile.

The improvement is real and survives the honest grid. It is also small (−1.85% event, −1.06% tick) and, as the next three subsections show, it is not a uniform improvement in any sense.

8.2 The gain is confined to one stratum

Table 9: RMSE of the learned model relative to persistence, by realised-movement tercile.

Tercile	Event grid	Tick grid
Low (quiescent)	+7.50% (worse)	+50.07% (worse)
Mid	−0.31%	+1.99% (worse)
High (volatile)	−2.25% (better)	−1.38% (better)

The aggregate win comes entirely from the volatile tercile. On quiescent markets the model is substantially worse in relative terms.

That +50% must be read carefully, and we state the correction ourselves. The quiescent stratum’s absolute RMSE is near zero (persistence 0.00763 → model 0.01148, a gap of 0.004), whereas the volatile stratum has a base of 0.162 and an absolute gain of 0.0022. Under a quadratic metric the volatile stratum dominates completely, so a 50% relative degradation on an almost-zero base contributes almost nothing to aggregate RMSE. *A large relative error on a tiny base is not a large error.* We initially read that +50% as a large exploitable inefficiency; Section 8.5 shows it is not.

8.3 Metric disagreement is the same finding

MAE does not improve on either grid. On the event grid the model reaches 0.05919 against persistence’s 0.05921 — a difference of 1.5×10^{-5} against a seed standard deviation of $\sim 4 \times 10^{-5}$, i.e. statistically indistinguishable and *not* an improvement. On the tick grid it is clearly worse (0.04550 vs 0.04488).

This is not a contradiction of Table 7; it is the same fact seen twice. RMSE is dominated by large errors, MAE by typical ones. A model that helps only on volatile markets necessarily improves the former and not the latter. The honest one-line summary:

The model reduces large errors on volatile markets while slightly degrading typical and quiet cases; aggregate RMSE improves, MAE and unconditional direction accuracy do not.

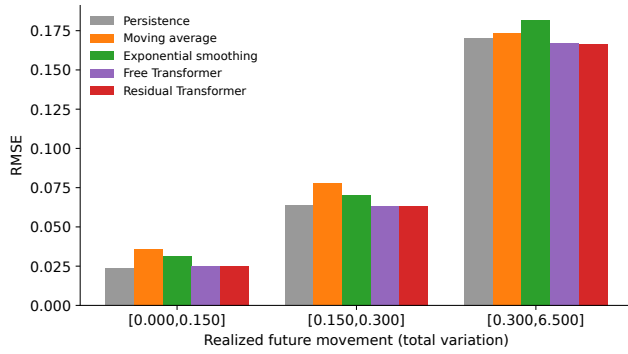


Figure 2: Error by realised-movement stratum. The learned model’s advantage is concentrated in the volatile stratum; in the quiescent stratum persistence is better in relative terms, on an almost-zero base.

Table 10: Learned model vs persistence, RMSE change by horizon step.

h	1	2	3	4	5	6	7	8	9	10
Event	-4.66	-2.70	-2.52	-1.72	-1.61	-1.82	-1.47	-1.37	-1.17	-0.95
Tick	-2.31	-1.20	-1.35	-1.17	-1.25	-1.20	-1.17	-0.92	-0.81	-0.75

8.4 Gains decay with horizon, contradicting our prior

We pre-registered the opposite expectation — that a learned model would help *more* at longer horizons, where persistence degrades. The measured effect decays monotonically on both grids. What we capture is **short-horizon microstructure**, not long-horizon dynamics. We report the direction we measured and note that it contradicted our prior.

8.5 A volatility-conditioned gate does not help

Since the gain is stratum-specific, the obvious next step is to route between persistence and the learned model on volatility observable at forecast time (mean absolute step over the last 10 observations), threshold selected on validation only. It gives **no significant RMSE improvement** on either grid: event 0.100950 gated against 0.100940 plain, tick 0.095212 against 0.095207.

Crucially this is not a tuning failure. An *oracle* gate conditioned on realised future movement — unattainable by construction — recovers only 0.09% and 0.14% (0.100855, 0.095078), so **no rule of this family has meaningful headroom**. That converts “we tried and it failed” into a stronger statement, and closes the most natural reviewer question with a measurement rather than speculation. One thing gating does buy is repairing the MAE regression: gated event MAE 0.055581 beats persistence’s 0.055605, where the plain model was worse at 0.055656. (These rows are from the pre-correction grids of Section 4.3; the experiment predates the fix, which has no mechanism to reverse a null.)

8.6 Ablation and reproducibility

The gain is not merely the persistence anchor. The non-residual Free Transformer, which predicts the path directly, also beats persistence significantly on all 3 seeds; the anchor helps (-0.00196 vs -0.00179) but is not the source of the effect.

The result replicates across two independent training tracks. The full protocol was executed end to end twice, once on Apple MPS and once on an NVIDIA A100, with independent evaluations — a stronger check than a re-run, since the tracks share no training process, RNG stream, or kernels. They agree on RMSE to 2.5×10^{-5} (event) and 6.6×10^{-5} (tick), inside the within-track seed standard deviation, and the gate passes independently on both.

8.7 The benchmark and the result replicate on a second venue

Every result above is Polymarket. To test whether any of it is platform-specific we rebuilt the entire benchmark on **Kalshi** — a different exchange, different market construction, different participants — using the identical pipeline, frozen configuration, and seed matrix.

Table 11: Cross-venue replication of the learned result, in units of 10^{-3} . Both columns are market-ID-split figures; see the magnitude caveat below.

($\times 10^{-3}$)	Polymarket	Kalshi
Event dRMSE vs persistence	-1.96	-2.63
95% CI	[-2.55, -1.38]	[-3.58, -1.68]
Tick dRMSE	-1.02	-2.20
95% CI	[-1.58, -0.50]	[-3.01, -1.38]
Free-forecaster ablation	-1.79	-2.46
Gate passes, both grids	yes	yes

The negatives replicate unchanged. Kalshi MAE remains worse than persistence on both grids (0.05803 vs 0.05742 event; 0.04460 vs 0.04351 tick), and tick direction accuracy remains worse (0.2929 vs 0.4222). Nothing improved by changing venue.

Direction, not magnitude. The Kalshi effect is nominally larger, and we deliberately decline to claim that. Section 6 shows Kalshi enters this comparison with roughly twice Polymarket’s event-family exposure (39.14% vs 19.86% on the event grid), so the two venues’ magnitudes are measured under different amounts of leakage and are not comparable on that axis. What is established is that *the direction of the replication holds on an independently constructed dataset*, for both the construction artifact and the learned result.

9 There is no tradeable edge, and we measure by how much

A forecasting-accuracy improvement is not an economic claim, and the two are routinely conflated. Testing an inefficiency net of transaction costs is the standard discipline [3], and it is what separates a detectable inefficiency from an exploitable one. We measure the edge in the units a trader would face, under assumptions chosen to be **generous to the strategy**: fills at the observed price, no

market impact, unlimited liquidity, the ability to act on every step, and perfect foreknowledge of which steps are non-flat. A negative result under these conditions is difficult to argue with.

Trade one unit in the predicted direction. Contracts settle in $[0, 1]$, so probability units are dollars per share. With directional accuracy q on steps that move and round-trip cost c ,

$$\mathbb{E}[\text{profit}] = (2q - 1) \mathbb{E}[|\Delta|] - c, \quad q^* = \frac{1}{2} + \frac{c}{2\mathbb{E}[|\Delta|]}.$$

Table 12: The edge priced in execution units.

	Event grid	Tick grid
Directional accuracy on tradeable steps	0.5200	0.5103
Gross P&L per trade	+0.00138	+0.00136
Max sustainable round-trip cost	0.00138	0.00136
as a fraction of the modal 0.0100 tick	0.14	0.14
Break-even accuracy vs a 1% spread	0.6671	0.6429
Shortfall	-14.7 pp	-13.3 pp

The entire tradeable edge is about one seventh of a single tick. The modal price increment is 0.0100, so the strategy cannot pay a seventh of the minimum increment before any spread, fee, or slippage is charged. Breaking even against a 1% spread would require roughly 64–67% directional accuracy against the 51–52% achieved.

One measurement runs in the strategy’s favour and we report it: correct calls land on larger moves than incorrect ones (1.026× event, 1.039× tick), so the edge is worth 1.33× and 1.94× what raw accuracy alone implies. This is consistent with the stratification result — the gain concentrates on volatile cases. It does not change the conclusion.

Both grids converge on the same answer, 0.14 ticks, so the economic negative is robust to the correction of Section 4.3.

The scientific claim — that these markets are not perfectly efficient tick-to-tick — stands. The economic claim does not, and the gap is now measured rather than asserted.

10 Limitations

The effect of family-disjoint splitting is unquantified. We measure exposure (19.86%–45.19% of test windows) and remove it (0.00%), but we do not report what removal costs the measured effect. A paired comparison requires both arms built by one query plan; we found our builder’s sort leaves ties unbroken, and since deduplication is adjacency-based and the window origin depends on post-deduplication length, two plans yield different windows for the same outcome. On Kalshi 27.34% of price rows share a timestamp with another row for the same market (Polymarket: 2.88%), so the ordering is load-bearing. The builder now derives both split vectors from one plan; the paired retrain is outstanding, and all model figures here are market-ID-split.

The residual bound binds exactly where the gains are. The head emits $0.25 \cdot \tanh(\cdot)$, so the model cannot deviate more than ± 0.25 from the anchor. That is what makes persistence a nested special case, but 32.4% of high-movement-tercile test trajectories contain at least one horizon point outside the reachable band (26.9% tick)

against 0.0% in the low-movement tercile; 3.9% of horizon points overall are unreachable. The consequence is favourable — the reported gain is a **lower bound** — but future work should widen or learn the scale.

Construction and scope. The sequence index is tick order, not wall-clock time, so “10 steps ahead” is a variable amount of real time. We use price history only — no news, order books, liquidity, or metadata — and one window per outcome, so the benchmark does not characterise every phase of a market’s life. Different markets can produce byte-identical near-degenerate paths: 0.13% of event-grid test trajectories (2.62% tick) have an identical twin in train, with a median of 1–2 distinct price levels. That is not a split defect, and its bias runs *against* the learned model, since persistence is exactly correct on a path that never moves; we disclose the rate and do not filter.

CUDA training is not bit-reproducible. We set determinism flags, but PyTorch’s memory-efficient attention backward remains non-deterministic and only warns. Independent re-runs landed within 8.4×10^{-6} (event) and 9.1×10^{-6} (tick) of the reported values, 5–8× inside the seed standard deviation, so the claims replicate but exact digits should not be expected to. Effect sizes are small in absolute terms throughout, which is why every comparison carries a paired-bootstrap interval and a per-seed gate.

11 Conclusion

Prediction-market probability paths are weakly predictable at short horizons in volatile markets, and not otherwise. The effect is statistically solid — two venues, two time grids, every seed’s interval excluding zero — and economically negligible, worth about a seventh of one tick.

The more useful finding is upstream of the model. A benchmark-construction step that reads as data hygiene silently disables the primary baseline and inflates its error by more than a quarter, and the standard direction metric carries no information about that baseline on either time indexing. Both are cheap to detect and cheap to report, and neither is visible unless someone looks.

What we do not claim. That prediction markets are solved; that the model predicts real-world outcomes better than the market does; that this performs counterfactual simulation or is an agentic world model; or that Transformers are novel for time series. Three further declines are worth stating explicitly. That persistence has directional skill on the tick grid — its 0.4477 is exactly the flat-step fraction and 0.0000 on every step that moves. That MAE improves — the event-grid difference is 1.5×10^{-5} against a seed standard deviation of $\sim 4 \times 10^{-5}$, which is indistinguishable. And that any of this is tradeable.

Future work. A market’s own history appears close to information-exhausted, so the natural next move is not a larger model but a *wider* one, conditioning on related markets. Both venues list **ladders**: families of contracts on one event at different thresholds, which supply a mechanical no-arbitrage constraint and therefore an evaluation axis baselines cannot satisfy by construction. That study is reported separately. Also outstanding: the paired retrain

quantifying the family-split effect (Section 10), wall-clock resampling, and scaling the benchmark from 100k to ~694k windows.

Ethics and Privacy Statement

This work uses only publicly quoted prices from resolved markets on two exchanges. No personal data, account identifiers, or individual trader activity is collected, and no human subjects are involved; the unit of analysis is a market outcome’s price series. The venues’ terms do not permit redistributing the raw feeds, so we release code, dataset manifests, the family-disjoint builder and parity-check rather than the underlying prices — enough to reproduce every number here from a licensed copy of the source.

The clearest foreseeable misuse of forecasting work on prediction markets is automated trading, and Section 9 addresses it directly: under assumptions deliberately generous to a trading strategy, the measured edge sustains roughly one seventh of a tick in round-trip cost, an order of magnitude short of viable. We report that null explicitly to discourage the inference that accuracy gains of this size are exploitable. The benefit we intend is methodological — a preprocessing convention that silently inflates a baseline causes published results to overstate model skill, and naming it with a number attached should reduce, not increase, the number of over-claimed results in this area.

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References

- [1] Joyce E. Berg, Forrest D. Nelson, and Thomas A. Rietz. 2008. Prediction Market Accuracy in the Long Run. *International Journal of Forecasting* 24, 2 (2008), 285–300.
- [2] Christoph Bergmeir and José M. Benítez. 2012. On the Use of Cross-Validation for Time Series Predictor Evaluation. *Information Sciences* 191 (2012), 192–213.
- [3] Eugene F. Fama. 1970. Efficient Capital Markets: A Review of Theory and Empirical Work. *The Journal of Finance* 25, 2 (1970), 383–417.
- [4] Shihao Gu, Bryan Kelly, and Dacheng Xiu. 2020. Empirical Asset Pricing via Machine Learning. *The Review of Financial Studies* 33, 5 (2020), 2223–2273.
- [5] Rob J. Hyndman and George Athanasopoulos. 2021. *Forecasting: Principles and Practice* (3rd ed.). OTexts, Melbourne, Australia.
- [6] Sayash Kapoor and Arvind Narayanan. 2023. Leakage and the Reproducibility Crisis in Machine-Learning-Based Science. *Patterns* 4, 9 (2023).
- [7] Shachar Kaufman, Saharon Rosset, Claudia Perlich, and Ori Stitelman. 2012. Leakage in Data Mining: Formulation, Detection, and Avoidance. *ACM Transactions on Knowledge Discovery from Data* 6, 4 (2012), 1–21.
- [8] Marcos López de Prado. 2018. *Advances in Financial Machine Learning*. John Wiley & Sons, Hoboken, NJ.
- [9] Spyros Makridakis, Evangelos Spiliotis, and Vassilios Assimakopoulos. 2020. The M4 Competition: 100,000 Time Series and 61 Forecasting Methods. *International Journal of Forecasting* 36, 1 (2020), 54–74.
- [10] Charles F. Manski. 2006. Interpreting the Predictions of Prediction Markets. *Economics Letters* 91, 3 (2006), 425–429.
- [11] Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. 2023. A Time Series is Worth 64 Words: Long-Term Forecasting with Transformers. In *International Conference on Learning Representations (ICLR)*.
- [12] David R. Roberts, Volker Bahn, Simone Ciuti, Mark S. Boyce, Jane Elith, Gurutzeta Guillera-Arroita, Severin Hauenstein, José J. Lahoz-Monfort, Boris Schröder, Wilfried Thuiller, David I. Warton, Brendan A. Wintle, Florian Hartig, and Carsten F. Dormann. 2017. Cross-Validation Strategies for Data with Temporal, Spatial, Hierarchical, or Phylogenetic Structure. *Ecography* 40, 8 (2017), 913–929.
- [13] David Rothschild. 2009. Forecasting Elections: Comparing Prediction Markets, Polls, and Their Biases. *Public Opinion Quarterly* 73, 5 (2009), 895–916.
- [14] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention Is All You Need. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- [15] Justin Wolfers and Eric Zitzewitz. 2004. Prediction Markets. *Journal of Economic Perspectives* 18, 2 (2004), 107–126.
- [16] Justin Wolfers and Eric Zitzewitz. 2006. *Interpreting Prediction Market Prices as Probabilities*. NBER Working Paper 12200. National Bureau of Economic Research.
- [17] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. 2023. Are Transformers Effective for Time Series Forecasting?. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- [18] Haoyi Zhou, Shanghang Zhang, Jieqi Peng, Shuai Zhang, Jianxin Li, Hui Xiong, and Wancai Zhang. 2021. Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*.